

Revision 05-07 • Attention Math + Transformer Architecture

Interactive revision of [lesson 05](#) and [lesson 07](#): each question as asked, the answer, where I slipped, and the one-line takeaway.

Q1 • Derive KV bytes per token, and what GQA buys SOLID

Covers: 05, 07, 19

QUESTION

Llama 3 8B: 32 layers, 8 KV heads, head dim 128, BF16. How many bytes of KV cache per token? Llama 2 7B is the same but with 32 KV heads (no GQA): what's its KV per token, and what does that do to users per GPU?

How I solved it (powers of two)

1. KV/token = 2 (K and V) $\times$ 32 (layers) $\times$ 8 (KV heads) $\times$ 128 (head dim) $\times$ 2 (bytes, BF16).
2. In powers of two: $2^1 \times 2^5 \times 2^3 \times 2^7 \times 2^1 = 2^{17} = 131,072$ bytes = **128 KiB**.
3. Llama 2 7B swaps 8 heads (2^3) for 32 (2^5): $2^{19} =$ **512 KiB**, 4 $\times$ more.
4. Same H100 and ~ 53 GB cache pool (lesson 18): 4 $\times$ the bytes per token means **4 $\times$ fewer tokens**, $\sim 100k$ instead of $\sim 400k$, so ~ 25 users at 4k context instead of ~ 100 .

RE-DRILL

Powers of two make these products instant: $32 = 2^5$, $8 = 2^3$, $128 = 2^7$, and $2^{10} = 1$ KiB.

Q2 • Parameters in one transformer block PARTIAL

Covers: 07, 12

QUESTION

Llama 3 8B: $h = 4,096$, MLP size $i = 14,336$, KV width $8 \times 128 = 1,024$. Q and O are $h \times h$; K and V are $h \times 1,024$; the MLP has three $h \times i$ matrices. How many parameters per block, and which part dominates?

Answer (asked for the worked answer; $h = 2^{12}$, KV width = 2^{10} , $i = 3.5 \times 2^{12}$)

Part	Matrices	Count
Q + O	$2 \times h \times h = 2 \times 2^{12} \times 2^{12} = 2^{25}$	$\approx 33.6\text{M}$
K + V	$2 \times h \times 1,024 = 2 \times 2^{12} \times 2^{10} = 2^{23}$	$\approx 8.4\text{M}$
Attention		$\approx 42\text{M}$
MLP (gate, up, down)	$3 \times h \times i = 3 \times 2^{12} \times 3.5 \times 2^{12} = 10.5 \times 2^{24}$	$\approx 176\text{M}$
Block		$\approx 218\text{M}$ (MLP $\approx 81\%$)

- Check: $32 \times 218\text{M} \approx 6.98\text{B}$, plus embedding + LM head $2 \times 128,256 \times 4,096 \approx 1.05\text{B} \rightarrow \mathbf{8.03\text{B}}$.
- **Why it matters:** decode time = bytes $\div$ bandwidth and the MLP holds $\sim 81\%$ of each block's bytes, so it's the first target for quantization and for MoE. Speeding up attention alone barely helps (Amdahl).

RE-DRILL

Re-drill the shape of each matrix: Q, O are $h \times h$; K, V are $h \times (\text{KV heads} \times \text{head dim})$; the MLP is three $h \times i$.

Q3 · Is MoE cheap to host? (Mixtral 8x7B) PARTIAL

Covers: 07, 18

QUESTION

"Mixtral 8x7B only uses $\sim 13\text{B}$ parameters per token, so host it like a 13B model on one 80 GB H100 in BF16." Mixtral stores 46.7B parameters, 12.9B active per token. Right or wrong, for memory and for compute?

Answer

- **Memory follows total parameters:** $46.7\text{B} \times 2 \text{ bytes} \approx \mathbf{93 \text{ GB}}$ in BF16 $> 80 \text{ GB}$, so **it doesn't fit**. Options: FP8 ($\sim 47 \text{ GB}$, leaving $\sim 72 - 47 - 3 \approx 22 \text{ GB}$ for KV cache) or 2 GPUs.
- **Compute follows active parameters:** $2 \times 12.9\text{B} \approx \mathbf{26 \text{ GFLOPs per token}}$, like a 13B model. That part is true.
- **Nuance:** at batch 1, decode reads only the active experts' weights, so it's fast like a 13B. In a big batch, tokens route to different experts, so nearly all 93 GB gets read each step and the advantage shrinks.

RE-DRILL

Always end a "can we?" question with the number and a yes/no.

Q4 · Why $\div \sqrt{d}$, and why subtract the max in softmax? PARTIAL

Covers: 05, 13

QUESTION

Attention computes $q \cdot k \div \sqrt{d}$ before softmax. What goes wrong without $\div \sqrt{d}$ ($d = 128$)?
 Softmax kernels subtract the row's maximum before e^x : why, and does it change the result?

Answer

- **$\div \sqrt{d}$:** a dot product of d random unit-variance numbers has variance d , so scores spread by $\sqrt{128} \approx 11$. Dividing by $\sqrt{d}$ brings the spread back to ~ 1 . Without it, softmax saturates into **nearly one-hot**: one token takes all the weight, attention stops blending, and gradients vanish in training.
- **Subtract the max:** prevents **overflow**: FP16 tops out at 65,504, and $e^{12} \approx 162,755$ already overflows.
- **Same result:** $e^{(x - m)} = e^x \cdot e^{-m}$, and the e^{-m} cancels between numerator and denominator. Only differences between scores matter (lesson 13).

RE-DRILL

For every "why do we do X?" give both halves: what breaks without it, and what it costs (here: nothing).